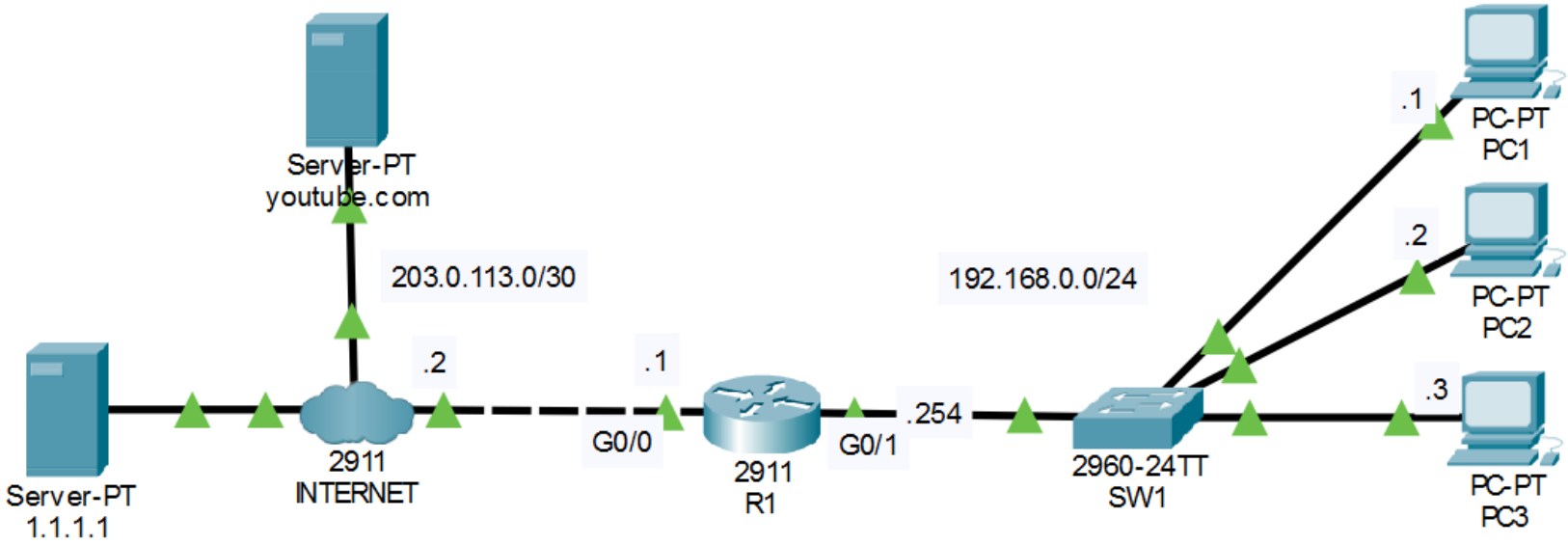
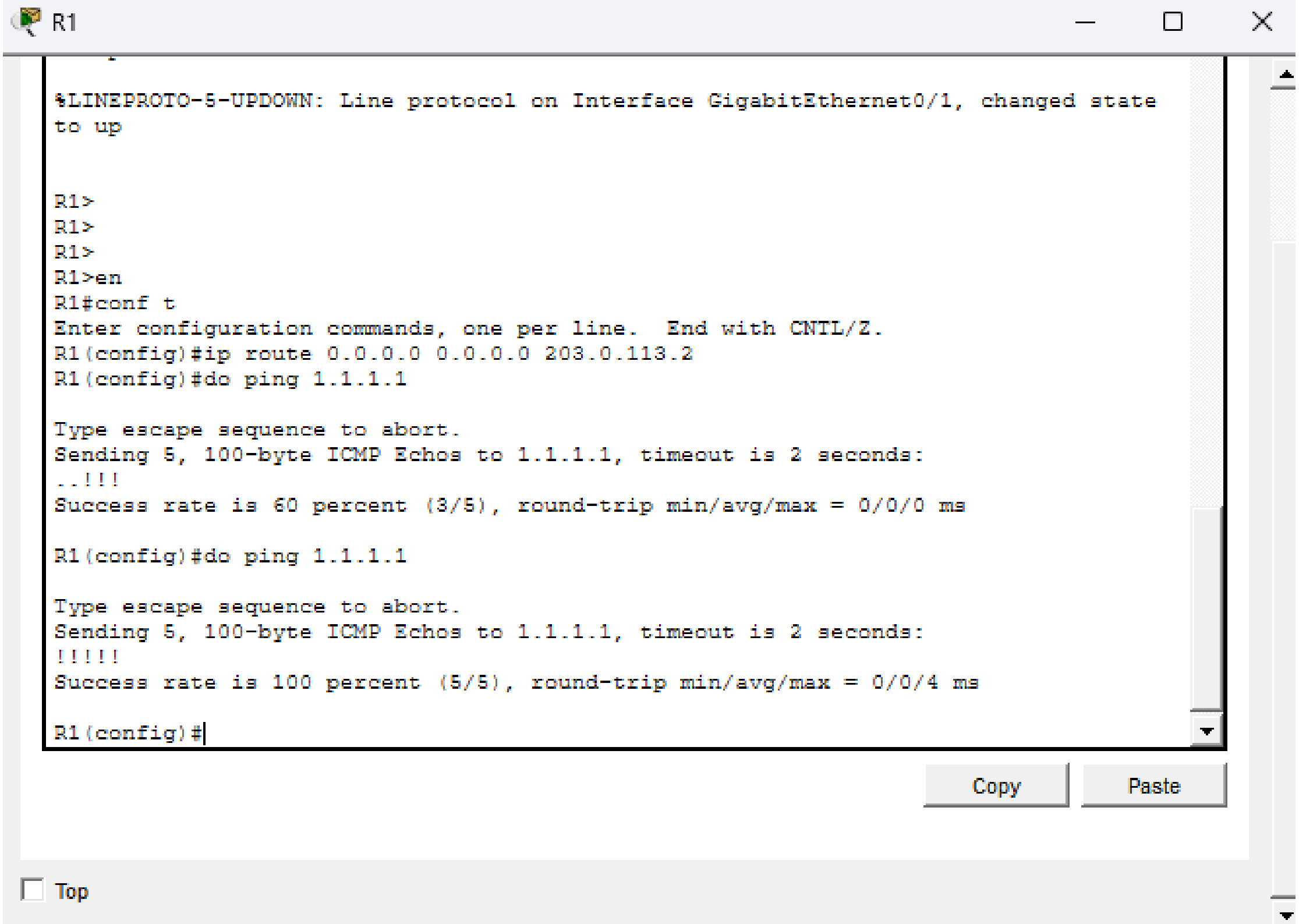


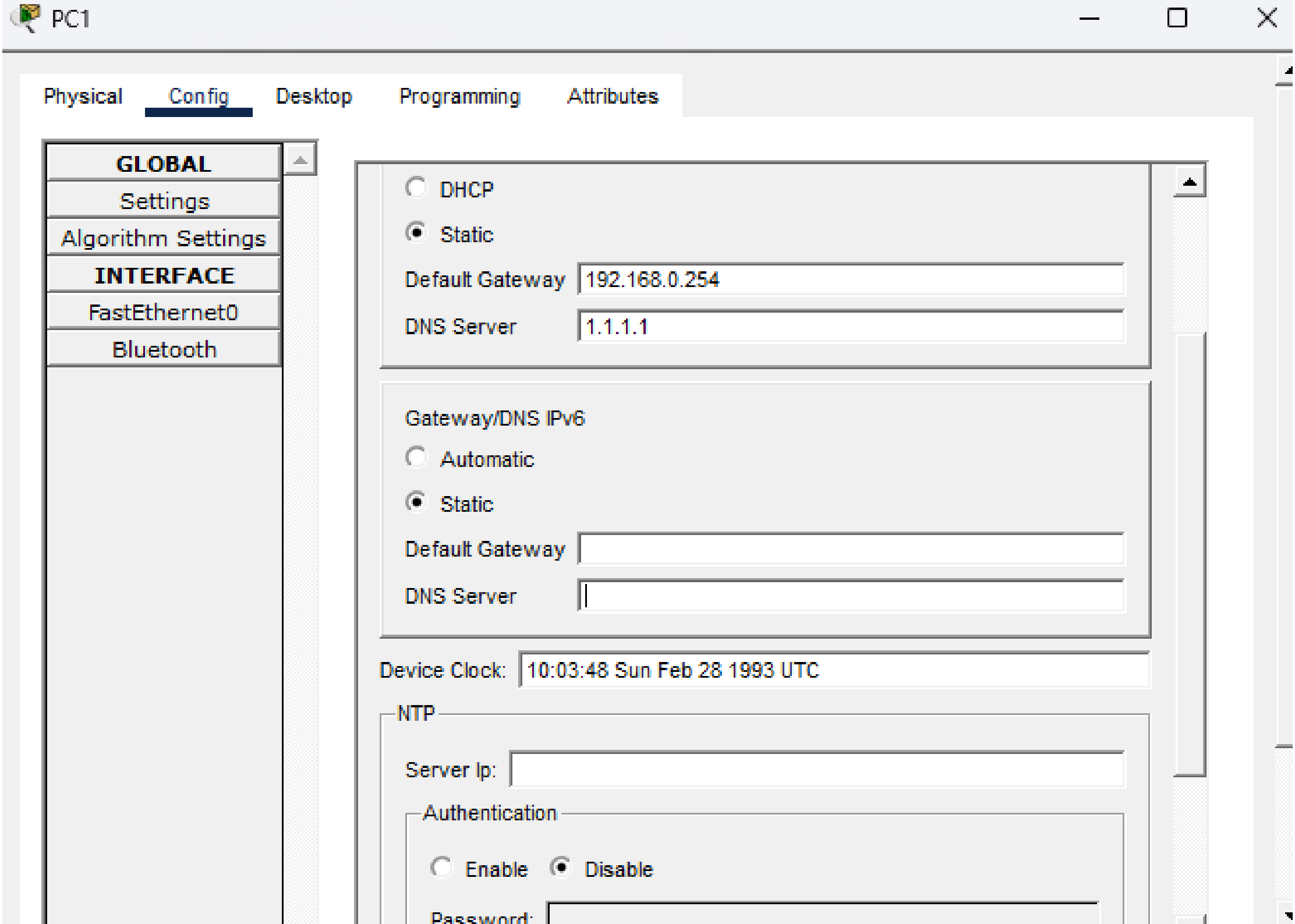
1. Configure a default route to the Internet on R1.

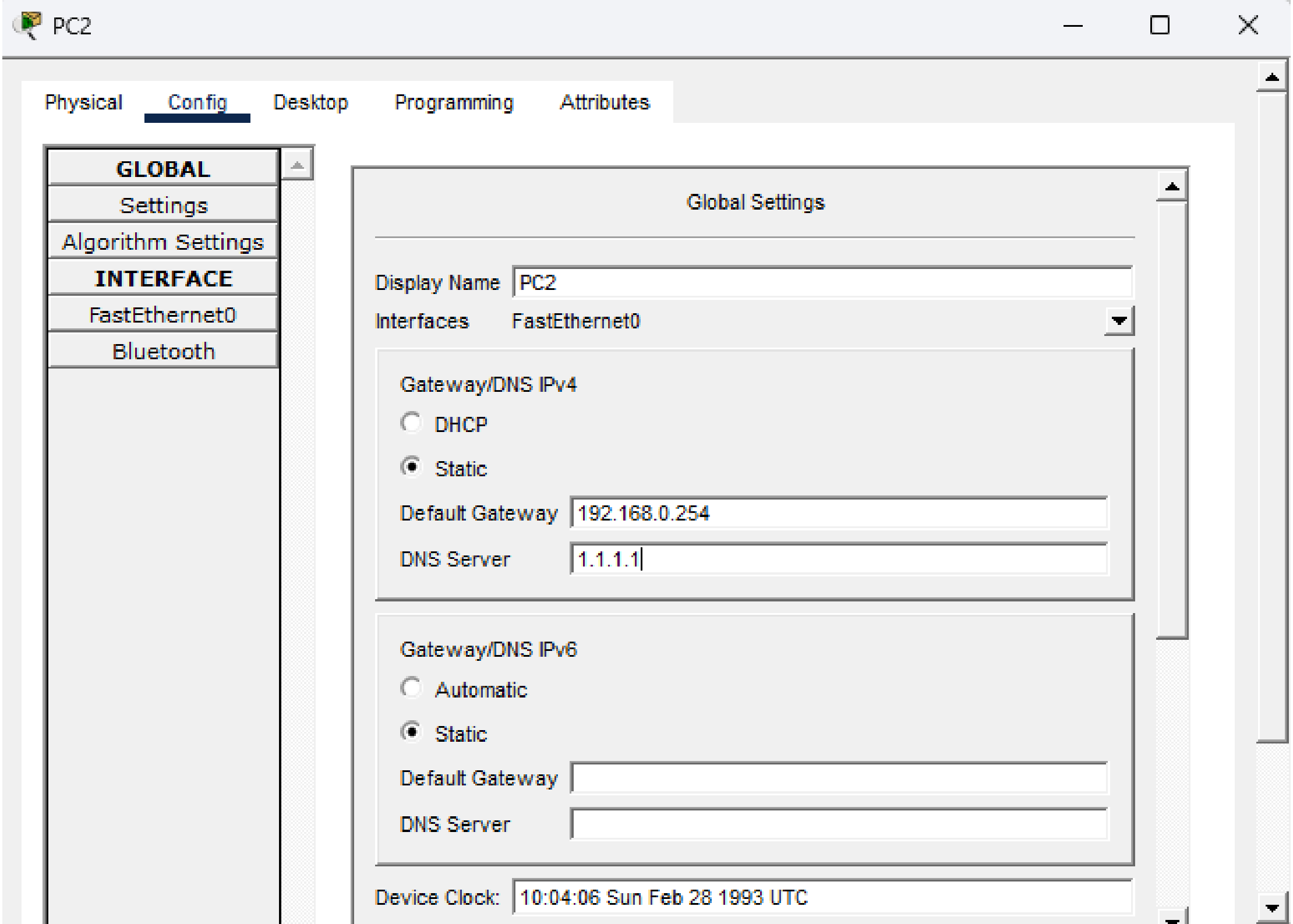
2. Configure PC1, PC2, and PC3 to use 1.1.1.1 as their DNS server.

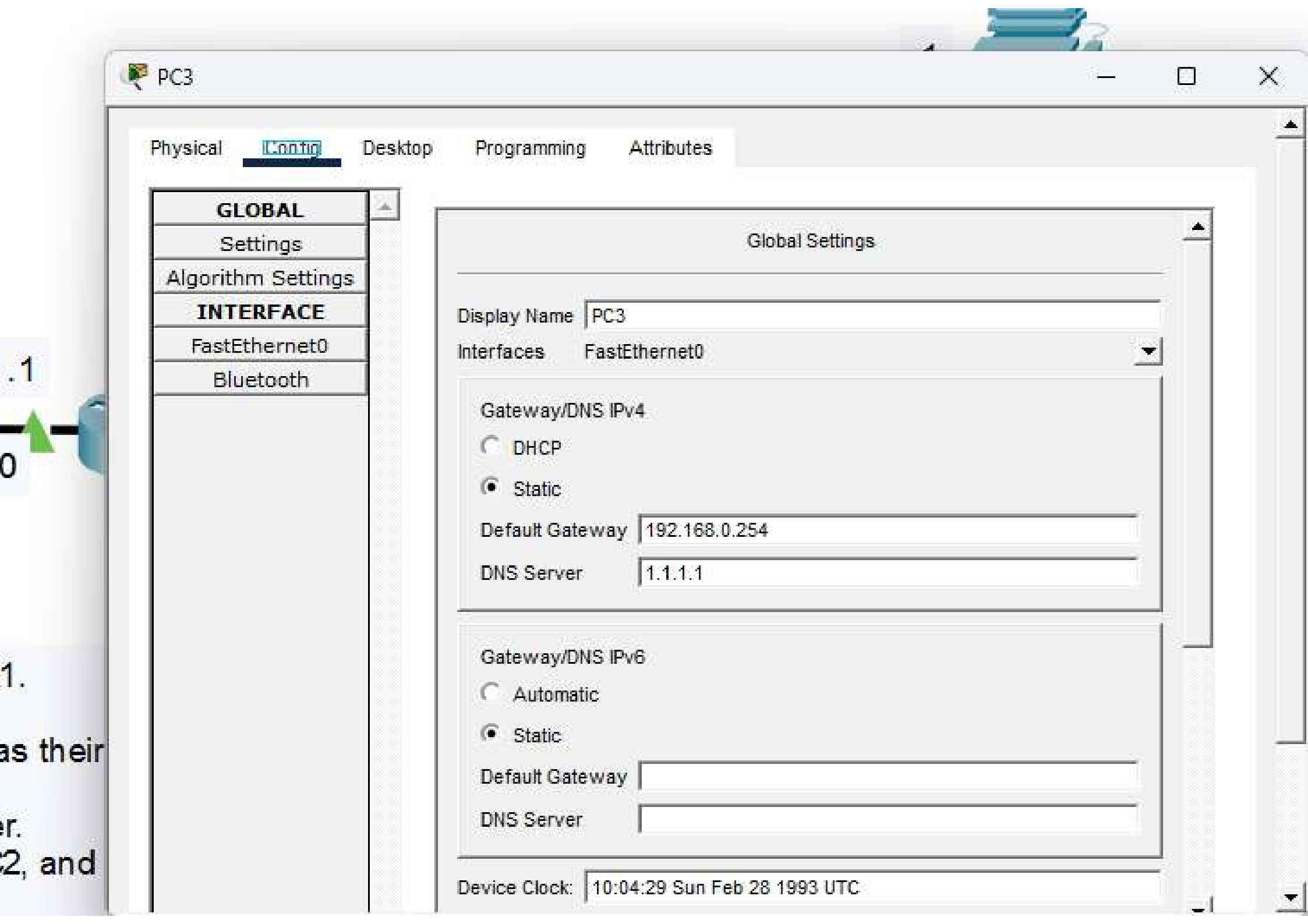
3. Configure R1 to use 1.1.1.1 as its DNS server.
   Configure host entries on R1 for R1, PC1, PC2, and PC3.
   Ping PC1 by name from R1.

4. #USE SIMULATION MODE FOR THIS STEP#
   From PC1, ping youtube.com by name. Analyze the messages being sent.


R1
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R1>
R1>
R1>
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1(config)#do ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms
R1(config)#do ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/4 ms
R1(config)#
Copy
Paste
Top


PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
DHCP
Static
Default Gateway
192.168.0.254
DNS Server
1.1.1.1
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock:
10:03:48 Sun Feb 28 1993 UTC
NTP
Server Ip:
Authentication
Enable
Disable
Password:


PC2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC2
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.0.254
DNS Server
1.1.1.1
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock:
10:04:06 Sun Feb 28 1993 UTC


PC3
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC3
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.0.254
DNS Server
1.1.1.1
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock:
10:04:29 Sun Feb 28 1993 UTC

```
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/4 ms

R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#ip name-server 1.1.1.1
R1(config)#ip host R1 192.168.0.254
R1(config)#ip host PC1 192.168.0.1
R1(config)#ip host PC2 192.168.0.2
R1(config)#ip host PC3 192.168.0.3
R1(config)#
R1(config)#do sh hosts
Default Domain is not set
Name/address lookup uses domain service
Name servers are 1.1.1.1

Codes: UN - unknown, EX - expired, OK - OK, ?? - revalidate
       temp - temporary, perm - permanent
       NA - Not Applicable None - Not defined

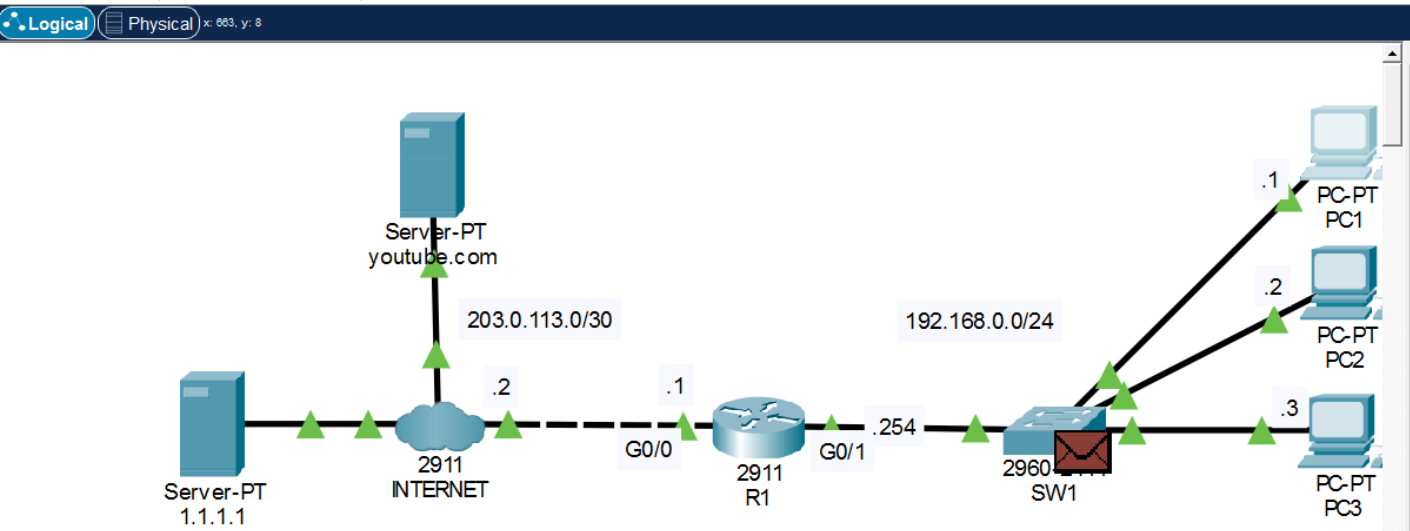
Host                      Port  Flags      Age Type   Address(es)
PC1                       None  (perm, OK)  0   IP      192.168.0.1
PC2                       None  (perm, OK)  0   IP      192.168.0.2
PC3                       None  (perm, OK)  0   IP      192.168.0.3
R1                        None  (perm, OK)  0   IP      192.168.0.254
R1(config)#do ping PC1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R1(config)#
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping youtube.com

Pinging 172.217.6.78 with 32 bytes of data:
```



1. Configure a default route to the Internet on R1.

2. Configure PC1, PC2, and PC3 to use 1.1.1.1 as their DNS server.

3. Configure R1 to use 1.1.1.1 as its DNS server.
   Configure host entries on R1 for R1, PC1, PC2, and PC3.
   Ping PC1 by name from R1.

4. #USE SIMULATION MODE FOR THIS STEP#
   From PC1, ping youtube.com by name. Analyze the messages being sent.

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC1	DNS
	0.001	PC1	SW1	DNS
	0.002	SW1	R1	DNS
	0.003	R1	INTERNET	DNS
	0.004	INTERNET	1.1.1.1	DNS
	0.005	1.1.1.1	INTERNET	DNS
	0.006	INTERNET	R1	DNS
	0.007	R1	SW1	DNS
	0.008	SW1	PC1	DNS
	0.008	--	PC1	ICMP
	0.009	PC1	SW1	ICMP
	0.010	SW1	R1	ICMP
	0.011	R1	INTERNET	ICMP
	0.011	--	INTERNET	ARP
	0.012	INTERNET	youtube.com	ARP
	0.013	youtube.com	INTERNET	ARP
	1.612	--	SW1	STP
	1.613	SW1	R1	STP
	1.613	SW1	PC1	STP
	1.613	SW1	PC2	STP
	1.613	SW1	PC3	STP
	3.612	--	SW1	STP
	3.613	SW1	R1	STP
	3.613	SW1	PC1	STP
	3.613	SW1	PC2	STP
	3.613	SW1	PC3	STP
	5.611	--	SW1	STP

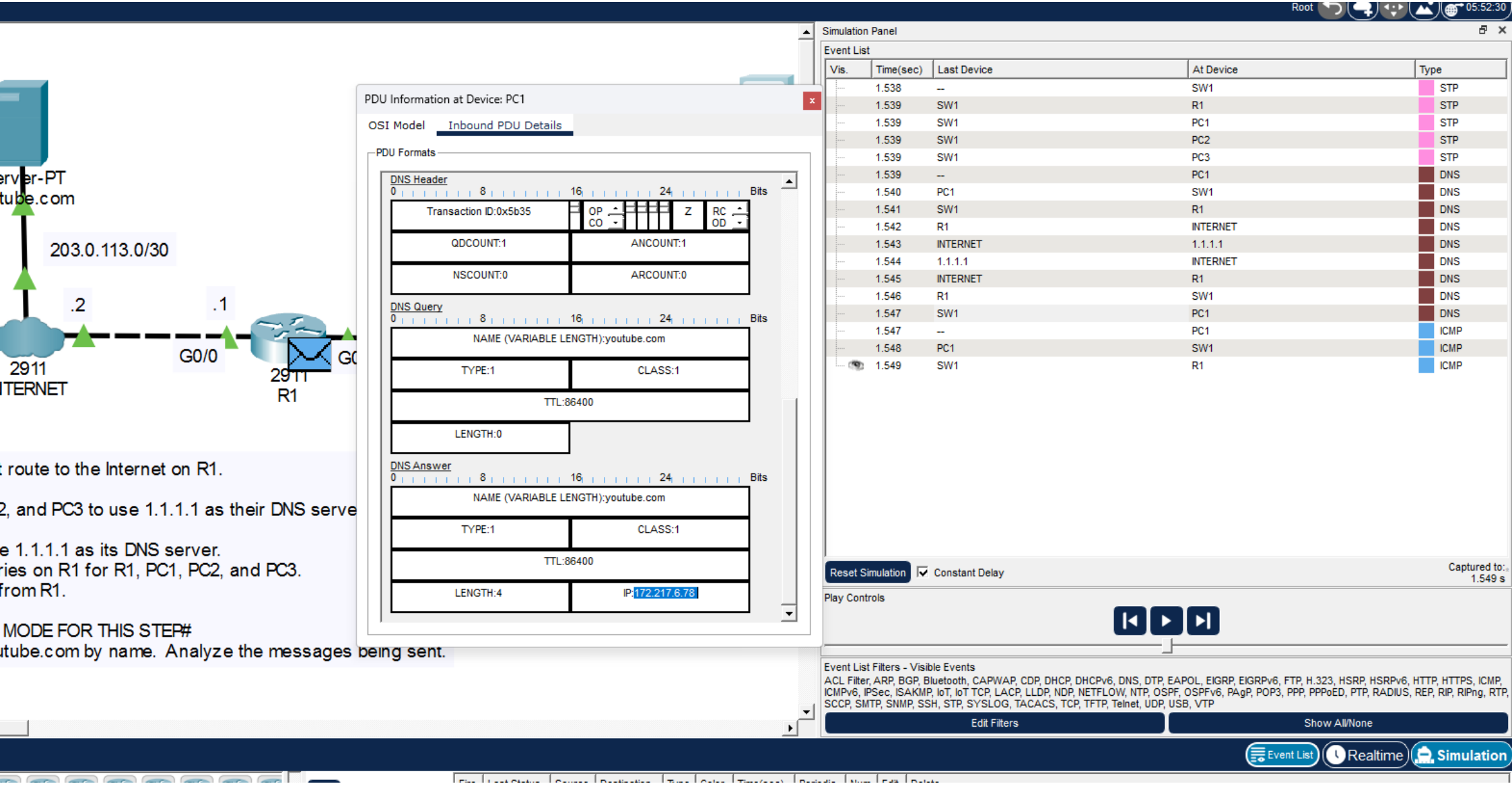
Reset Simulation | Constant Delay | Captured to: 6.016 s

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters | Show All/None


Root
05:52:30
erver-PT
tube.com
203.0.113.0/30
.2
.1
G0/0
2911
ITERNET
2911
R1
t route to the Internet on R1.
2, and PC3 to use 1.1.1.1 as their DNS serve
e 1.1.1.1 as its DNS server.
ries on R1 for R1, PC1, PC2, and PC3.
from R1.
MODE FOR THIS STEP#
utube.com by name. Analyze the messages being sent.
PDU Information at Device: PC1
OSI Model
Inbound PDU Details
PDU Formats
DNS Header
0 8 16 24 Bits
Transaction ID:0x5b35
OP
CO
Z
RC
OD
QDCOUNT:1
ANCOUNT:1
NSCOUNT:0
ARCOUNT:0
DNS Query
0 8 16 24 Bits
NAME (VARIABLE LENGTH):youtube.com
TYPE:1
CLASS:1
TTL:86400
LENGTH:0
DNS Answer
0 8 16 24 Bits
NAME (VARIABLE LENGTH):youtube.com
TYPE:1
CLASS:1
TTL:86400
LENGTH:4
IP:172.217.6.78
Simulation Panel
Event List
Vis. Time(sec) Last Device At Device Type
1.538 -- SW1 STP
1.539 SW1 R1 STP
1.539 SW1 PC1 STP
1.539 SW1 PC2 STP
1.539 SW1 PC3 STP
1.539 -- PC1 DNS
1.540 PC1 SW1 DNS
1.541 SW1 R1 DNS
1.542 R1 INTERNET DNS
1.543 INTERNET 1.1.1.1 DNS
1.544 1.1.1.1 INTERNET DNS
1.545 INTERNET R1 DNS
1.546 R1 SW1 DNS
1.547 SW1 PC1 DNS
1.547 -- PC1 ICMP
1.548 PC1 SW1 ICMP
1.549 SW1 R1 ICMP
Reset Simulation
Constant Delay
Captured to: 1.549 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Event List
Realtime
Simulation